1 Gain law

$(e_\theta = \theta - \hat{\theta})$

$$\bar{a}(\bar{w}) = 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)}$$
$$\bar{a}'(\bar{a}, b) := \frac{d\bar{a}}{d\bar{w}} = b\big(1 - \bar{a}\big)^2$$

2 Control law

$(d = 2)$

$$\delta\bar{w}_3[k] = \delta\bar{w}_1[k] + \sum_{i=1}^{d} \Delta\bar{w}_d[k-i] - \sum_{i=1}^{d} \bar{a}_w[k-i]\big\{\bar{f}_{dw}[k-i] + \bar{f}_T[k-i]\big\}$$

$$\boxed{\bar{f}_{dw}[k] = \hat{\bar{a}}_w^{-1}[k]\Big\{\Delta\bar{w}_d[k] + (1-\lambda_c)\sum_{i=1}^{d}\Delta\bar{w}_d[k-i] + (1-\lambda_c)\Big[y_1[k] - \sum_{i=1}^{d}\hat{\bar{a}}_w[k-i]\,\bar{f}_{dw}[k-i]\Big]\Big\}}$$

$$y_1[k] = \delta\bar{w}_1[k] + n_w[k]$$

3 True dynamics

$$\bar{w}[k] = \bar{w}_d[k] - \delta\bar{w}_3[k]$$
$$\Delta\bar{w}[k] := \bar{w}[k+1] - \bar{w}[k]$$
$$e_{\bar{a}_w}[k-i] := \bar{a}_w[k-i] - \hat{\bar{a}}_w[k-i]$$

$$\delta\bar{w}_3[k+1] = \delta\bar{w}_3[k] + \Delta\bar{w}_d[k] - \bar{a}_w[k]\big\{\bar{f}_{dw}[k] + \bar{f}_T[k]\big\} \tag{1}$$

$$\bar{a}_w[k]\,\bar{f}_{dw}[k] = \hat{\bar{a}}_w[k]\,\bar{f}_{dw}[k] + e_{\bar{a}_w}[k]\,\bar{f}_{dw}[k] \tag{2}$$

$$\begin{aligned}\hat{\bar{a}}_w[k]\,\bar{f}_{dw}[k] = {} & \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + (1-\lambda_c)\sum_{i=1}^{d} e_{\bar{a}_w}[k-i]\,\bar{f}_{dw}[k-i] \\ & + (1-\lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]\end{aligned} \tag{3}$$

$$\delta\bar{w}_3[k+1] = \lambda_c\,\delta\bar{w}_3[k] - \bar{f}_{dw}[k]\,e_{\bar{a}_w}[k] - (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i]\,e_{\bar{a}_w}[k-i] - \varepsilon_w[k] \tag{4}$$

$$\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]$$
$$\bar{a}_w[k+1] = \bar{a}_w[k] + b\big(1-\bar{a}_w[k]\big)^2\,\Delta\bar{w}[k]$$
$$b[k+1] = b[k]$$

4 Estimator

$$\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b\,\big]^\top$$

$$\delta\hat{\bar{w}}_1[k+1] = \delta\hat{\bar{w}}_2[k]$$
$$\delta\hat{\bar{w}}_2[k+1] = \delta\hat{\bar{w}}_3[k]$$
$$\delta\hat{\bar{w}}_3[k+1] = \lambda_c\,\delta\hat{\bar{w}}_3[k]$$
$$\Delta\hat{\bar{w}}[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]$$
$$\hat{\bar{a}}_w[k+1] = \hat{\bar{a}}_w[k] + \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2\,\Delta\hat{\bar{w}}[k]$$
$$\hat{b}[k+1] = \hat{b}[k]$$

5 Error dynamics

$$e_{\Delta\bar{w}}[k] := \Delta\bar{w}[k] - \Delta\hat{\bar{w}}[k]$$

$$e_{\delta\bar{w}_1}[k+1] = e_{\delta\bar{w}_2}[k]$$
$$e_{\delta\bar{w}_2}[k+1] = e_{\delta\bar{w}_3}[k]$$
$$e_{\delta\bar{w}_3}[k+1] = e_{\delta\bar{w}_3}[k] - e_{\Delta\bar{w}}[k] \tag{5}$$
$$e_{\bar{a}_w}[k+1] = \Big(1-2\hat{b}\big(1-\hat{\bar{a}}_w[k]\big)\Delta\hat{\bar{w}}[k]\Big)e_{\bar{a}_w}[k] + \hat{b}\big(1-\hat{\bar{a}}_w[k]\big)^2 e_{\Delta\bar{w}}[k] + \big(1-\hat{\bar{a}}_w[k]\big)^2\Delta\hat{\bar{w}}[k]\,e_b[k] \tag{6}$$
$$e_b[k+1] = e_b[k]$$

$$e_{\Delta\bar{w}}[k] = (1-\lambda_c)\,e_{\delta\bar{w}_3}[k] + \bar{f}_{dw}[k]\,e_{\bar{a}_w}[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i]\,e_{\bar{a}_w}[k-i] + \varepsilon_w[k] \tag{7}$$

(A)

$$
\begin{aligned}
e_{\bar{a}_w}[k-i] &= e_{\bar{a}_w}[k] \\
e_{\Delta\bar{w}}[k] &= (1-\lambda_c)\, e_{\delta\bar{w}_3}[k] + F_{dw}[k]\, e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
F_{dw}[k] &= \bar{f}_{dw}[k] + (1-\lambda_c) \sum_{i=1}^{d} \bar{f}_{dw}[k-i]
\end{aligned}
$$

(B)

$$
\begin{aligned}
e_{\Delta\bar{w}}[k-1] &= e_{\delta\bar{w}_2}[k] - e_{\delta\bar{w}_3}[k] \\
e_{\Delta\bar{w}}[k-2] &= e_{\delta\bar{w}_1}[k] - e_{\delta\bar{w}_2}[k]
\end{aligned}
$$

$$
e_{\bar{a}_w}[k-1] = \frac{e_{\bar{a}_w}[k] - \hat{b}\big(1-\hat{\bar{a}}_w[k-1]\big)^2\big(e_{\delta\bar{w}_2}[k] - e_{\delta\bar{w}_3}[k]\big) - \big(1-\hat{\bar{a}}_w[k-1]\big)^2 \Delta\hat{\bar{w}}[k-1]\, e_b[k]}{1 - 2\hat{b}\big(1-\hat{\bar{a}}_w[k-1]\big)\Delta\hat{\bar{w}}[k-1]} \tag{8}
$$

$$
e_{\bar{a}_w}[k-2] = \frac{e_{\bar{a}_w}[k-1] - \hat{b}\big(1-\hat{\bar{a}}_w[k-2]\big)^2\big(e_{\delta\bar{w}_1}[k] - e_{\delta\bar{w}_2}[k]\big) - \big(1-\hat{\bar{a}}_w[k-2]\big)^2 \Delta\hat{\bar{w}}[k-2]\, e_b[k]}{1 - 2\hat{b}\big(1-\hat{\bar{a}}_w[k-2]\big)\Delta\hat{\bar{w}}[k-2]} \tag{9}
$$

(B′)

$$
e_{\bar{a}_w}[k-1] = e_{\bar{a}_w}[k] - \hat{b}\big(1-\hat{\bar{a}}_w[k]\big)^2\big(e_{\delta\bar{w}_2}[k] - e_{\delta\bar{w}_3}[k]\big) - \big(1-\hat{\bar{a}}_w[k]\big)^2 \Delta\hat{\bar{w}}[k-1]\, e_b[k] \tag{8′}
$$

$$
\begin{aligned}
e_{\bar{a}_w}[k-2] = e_{\bar{a}_w}[k] &- \hat{b}\big(1-\hat{\bar{a}}_w[k]\big)^2\big(e_{\delta\bar{w}_1}[k] - e_{\delta\bar{w}_3}[k]\big) \\
&- \big(1-\hat{\bar{a}}_w[k]\big)^2\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big) e_b[k]
\end{aligned} \tag{9′}
$$

$$
\begin{aligned}
e_{\Delta\bar{w}}[k] = (1-\lambda_c)\, & e_{\delta\bar{w}_3}[k] + F_{dw}[k]\, e_{\bar{a}_w}[k] \\
&- (1-\lambda_c)\, \hat{b}\big(1-\hat{\bar{a}}_w[k]\big)^2\Big(\bar{f}_{dw}[k-1]\big(e_{\delta\bar{w}_2}[k] - e_{\delta\bar{w}_3}[k]\big) + \bar{f}_{dw}[k-2]\big(e_{\delta\bar{w}_1}[k] - e_{\delta\bar{w}_3}[k]\big)\Big) \\
&- (1-\lambda_c)\big(1-\hat{\bar{a}}_w[k]\big)^2\Big(\bar{f}_{dw}[k-1]\, \Delta\hat{\bar{w}}[k-1] + \bar{f}_{dw}[k-2]\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big)\Big) e_b[k] \\
&+ \varepsilon_w[k]
\end{aligned}
$$

F_e

$$F_e^{(A)}[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 & 1+\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} \quad -2\hat{b}\big(1-\hat{\bar{a}}_w\big)\Delta\hat{\bar{w}} & \big(1-\hat{\bar{a}}_w\big)^2\Delta\hat{\bar{w}} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned}
F_e^{(B')}[k]_{1,:} &= F_e^{(A)}[k]_{1,:} \\
F_e^{(B')}[k]_{2,:} &= F_e^{(A)}[k]_{2,:} \\
F_e^{(B')}[k]_{5,:} &= F_e^{(A)}[k]_{5,:}
\end{aligned}$$

$$\begin{aligned}
F_e^{(B')}[k]_{31} &= (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 \bar{f}_{dw}[k-2] \\
F_e^{(B')}[k]_{32} &= (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 \bar{f}_{dw}[k-1] \\
F_e^{(B')}[k]_{33} &= \lambda_c - (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2\big(\bar{f}_{dw}[k-1] + \bar{f}_{dw}[k-2]\big) \\
F_e^{(B')}[k]_{34} &= -F_{dw} \\
F_e^{(B')}[k]_{35} &= (1-\lambda_c)\big(1-\hat{\bar{a}}_w\big)^2\Big(\bar{f}_{dw}[k-1]\,\Delta\hat{\bar{w}}[k-1] + \bar{f}_{dw}[k-2]\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big)\Big)
\end{aligned}$$

$$\begin{aligned}
F_e^{(B')}[k]_{41} &= -(1-\lambda_c)\,\hat{b}^2\big(1-\hat{\bar{a}}_w\big)^4 \bar{f}_{dw}[k-2] \\
F_e^{(B')}[k]_{42} &= -(1-\lambda_c)\,\hat{b}^2\big(1-\hat{\bar{a}}_w\big)^4 \bar{f}_{dw}[k-1] \\
F_e^{(B')}[k]_{43} &= (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2\Big(1+\hat{b}\big(1-\hat{\bar{a}}_w\big)^2\big(\bar{f}_{dw}[k-1] + \bar{f}_{dw}[k-2]\big)\Big) \\
F_e^{(B')}[k]_{44} &= 1+\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} - 2\hat{b}\big(1-\hat{\bar{a}}_w\big)\Delta\hat{\bar{w}} \\
F_e^{(B')}[k]_{45} &= \big(1-\hat{\bar{a}}_w\big)^2\Delta\hat{\bar{w}} \\
&\quad - (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^4\Big(\bar{f}_{dw}[k-1]\,\Delta\hat{\bar{w}}[k-1] + \bar{f}_{dw}[k-2]\big(\Delta\hat{\bar{w}}[k-1] + \Delta\hat{\bar{w}}[k-2]\big)\Big)
\end{aligned}$$

$$\mathbf{q}^{(A)}[k] = \mathbf{q}^{(B')}[k] = \Big[\,0,\ 0,\ -\varepsilon_w[k],\ \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\varepsilon_w[k],\ 0\,\Big]^\top$$